Malicious PDF Detection using Machine Unlearning

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***Abstract*—The Portable Document Format (PDF) is a significant source of malware distribution due to its widespread use and inherent complexity. Machine Learning (ML)-based detectors offer a powerful defence against new and disguised threats; however, their adaptive nature makes them vulnerable to training data poisoning, also known as causative attacks. These attacks involve adversaries intentionally corrupting a model's training data, leading to a persistent decline in accuracy. The traditional solution of retraining the entire model from scratch on a clean dataset is often computationally expensive and creates a period of vulnerability. This paper proposes machine unlearning as an efficient and robust solution to this problem. Machine unlearning enables a model to selectively "forget" specific training samples, thereby removing the influence of poisoned data without the high costs associated with full retraining. This summary explores the intersection of PDF malware detection and machine unlearning, reviewing current PDF malware detection techniques and key machine unlearning methodologies. We introduce a conceptual framework that applies unlearning to repair and strengthen ML-based PDF malware detectors, enhancing their security and resilience in adversarial environments.**

***Keywords—Malicious* PDF Detection, Machine Unlearning, Model Repair, One-Class SVM, PJ SCAN. Malware Detection, SISA Training, Poisoning Attacks**

NOMENCLATURE

Nomenclature	Referred to
PDF	Portable Document Format
DBSCAN	Density-Based Spatial Clustering of Application with Noise
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
M-DBSCAN:	Modified Density-Based Spatial Clustering of Application with Noise
BERT	Bidirectional Encoder Representations from Transformer

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I. Introduction

As essential tools for digital communication, PDF documents are frequently targeted by cybercriminals because they can embed various scripts and content. Traditional signature-based antivirus solutions are often ineffective against modern polymorphic malware, which can easily bypass them through minor changes to their payload. Therefore, there has been a widespread adoption of machine learning-based systems that analyze intrinsic file characteristics to distinguish between benign and malicious files.

The integration of machine learning (ML) into security systems introduce new vulnerabilities, as adversaries can now compromise these learning-based systems by corrupting the training data itself, rather than solely focusing on evading the final model. This presents a significant challenge: how can a system operator efficiently restore a detector's integrity after identifying poisoned data in a training set? The traditional method of discarding the current model and retraining it with a clean dataset is inefficient, potentially taking days for complex models and leaving the system exposed during this period.

This paper proposes machine unlearning as an effective solution. Machine unlearning is a method for programmatically eliminating the influence of specific data points from a trained model, essentially causing it to "forget" them. The objective is to achieve a model state that is virtually identical to one retrained from scratch, but in a significantly shorter timeframe. Applying unlearning techniques to PDF malware detectors can bolster their security and reliability by enabling swift recovery from poisoning attacks.

The inherent vulnerabilities of the models used in these systems exacerbate the problem. PDF malware detectors frequently employ classical ML algorithms, such as Support Vector Machines, which are trained on features derived from static analysis to detect suspicious structural elements and embedded code patterns, including malicious JavaScript. While robust feature sets are essential for effectiveness, cunning adversaries can leverage this knowledge to launch both poisoning (causative) and evasion (exploratory) attacks. In an evasion attack, a malicious document is engineered to be misclassified by the deployed model. Beyond facilitating security recovery, machine unlearning is also crucial for regulatory compliance. It allows systems to adhere to the "Right to be Forgotten" mandates of data privacy regulations like GDPR and CCPA by programmatically removing the impact of specific user data or model-identifying attributes, thereby maintaining system integrity and trustworthiness.

II. Related Work

Our work connects two different fields of study: first, the methods used to find harmful PDF files, and second, the growing need for what's called "machine unlearning."

A. Vulnerabilities and *Advances in malicious PDF detection*

Identifying harmful PDF files continues to be a vital and intensely researched topic in cybersecurity. The vast number and increasing sophistication of threats have driven a shift from traditional signature-based approaches to more adaptive, learning-oriented systems. As a result, machine learning classifiers have become central to contemporary detection process

Extensive reviews of the field reveal that supervised algorithms, especially ensemble techniques like Random Forest and Boosting, kernel-based approaches such as Support Vector Machines (SVMs), and probabilistic models like Bayesian classifiers, have been effective in this domain [6]. Nonetheless, this effectiveness is challenged by an ongoing adversarial relationship. These systems are key targets for attackers who craft advanced evasion strategies, often by making minor changes to malicious files that can deceive a classifier. This ongoing "cat-and-mouse game" highlights the necessity for constant advancement in detection strategies [1].

Research has primarily concentrated on improving model performance by employing sophisticated feature engineering techniques. The underlying idea is that more detailed and resilient features can reduce a model's vulnerability to basic evasion strategies. A notable study enhanced a typical set of 29 structural features by introducing six innovative "anomaly-based" features. These new features were crafted to identify irregularities in file structures, such as discrepancies between obj and end obj tags or the suspicious simultaneous presence of embedded files and JavaScript code. Incorporating these features significantly boosted detection rates in five out of the seven classifiers evaluated, with a Random Forest model achieving an impressive accuracy of 99.65% on the Evasive-PDFMal2022 dataset. This underscores the importance of input feature quality, which is as crucial as the selection of the learning algorithm itself [7].

In addition to selecting features, the design of the detector's architecture is crucial. Static analysis, which involves examining a file's structure and content without executing or rendering it, is a preferred method because it avoids the risks and resource demands of dynamic analysis in a sandbox environment. PJScan is a notable example of a static analysis system. It focuses on the lexical analysis of embedded JavaScript, a frequent target for exploits. To construct its classifier, PJScan uses a technology stack that includes the Poppler library for reliable script extraction and a modified version of Mozilla's SpiderMonkey JavaScript interpreter for safe code tokenization. Given the challenge of acquiring a large, accurately labeled dataset of benign PDFs containing JavaScript, PJScan utilizes a One-Class SVM (OCSVM). This model is trained solely on malicious samples to define a precise boundary around malware characteristics, classifying any new sample outside this boundary as benign [8]. The effectiveness of this one-class approach is supported by foundational research in document classification, which indicates that OCSVMs perform exceptionally well when positive-class examples are plentiful but negative-class examples are limited or absent. However, this research also warns that OCSVM performance can be highly sensitive to parameter adjustments, especially regarding data representation and the choice of kernel function [9].

B. *The Emergence Of Machine Unlearning*

We need a way to correct the model when data poisoning attacks are successful or when we just need to delete old data for privacy reasons. The expanding field of "machine unlearning" seeks to address this fundamental issue.

One of the most promising frameworks for general use achieves the goal of unlearning by transforming machine learning algorithms into a sum of individual statistical contributions for each sample. This transformation enables the efficient removal of a data point through a subtraction operation. The effectiveness of this method has been proven by its successful implementation in four varied, real-world systems, such as a malware detector (Zozzle), a recommendation engine (LensKit), and notably, the PJScan system, which establishes a clear connection between unlearning theory and security applications [2]. Beyond these general techniques, highly specialized solutions have been crafted for specific algorithms that are widely used in security. For example, the DYNFRS framework is specifically designed for Random Forest classifiers. It incorporates a distinctive subsampling algorithm to minimize the impact of any single sample, along with a lazy tag strategy (LZY). This strategy defers the immediate reconstruction of trees by temporarily marking affected leaf nodes as invalid, delaying the computational effort until a prediction actually requires that node, thus optimizing efficiency [3].

A taxonomy of unlearning techniques is provided by surveys in this field, and they can be broadly divided into a number of different approaches [6]. A model-agnostic approach that reframes the learning algorithm into a summation form is one of the most flexible. This makes it possible to "forget" data using a straightforward subtraction operation, which is asymptotically quicker than retraining. Crucially, this approach has a direct connection to the security domain because it has been verified on actual systems, such as the PJScan detector [7]. Specialized frameworks have been created for particular models in addition to these general-purpose techniques. For instance, by employing a lazy-update approach to handle data removal requests, the DYNFRS framework offers an effective unlearning solution suited to Random Forest classifiers, which are frequently used in malware detection [8].

C. *Bridging the Gap: Unlearning for Security*

A thorough analysis of the literature indicates a notable discrepancy between the fields of machine learning and malicious content detection research. Despite their maturity, the two fields have developed mainly simultaneously. While the unlearning community has concentrated on the effectiveness and verifiability of data removal, the security community has concentrated on creating extremely accurate detectors. Because of this, even the most advanced malware detection systems, such as PJScan [8] or those that use enhanced feature sets [7], are not always resistant to compromises of training data.

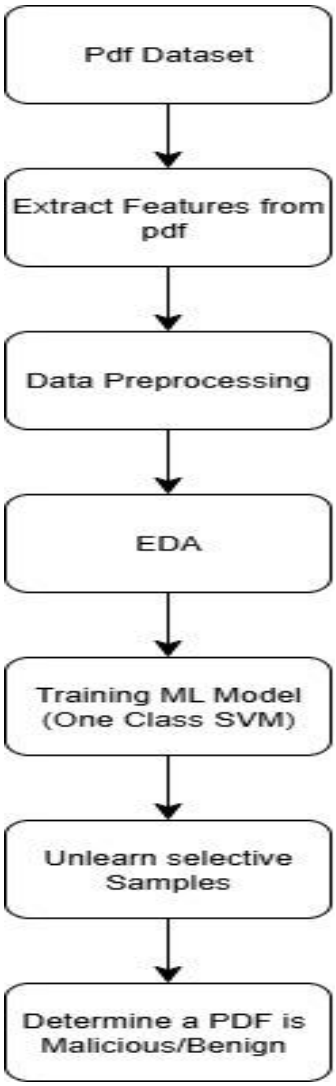
This discrepancy poses a serious risk. As explained in [4] and [5], an adversary could successfully carry out a data poisoning attack, gradually tainting the detector's model. The system operator's only option, in the absence of an effective remediation mechanism, would be to discard the compromised model and begin a thorough retraining procedure. In addition to being computationally prohibitive, this would leave the system vulnerable or offline for an extended amount of time.

The goal of this work is to close that crucial gap. We contend that security systems need to be easily correctable in addition to accurate in order to be operationally resilient. A new class of security systems that can gracefully recover from data poisoning attacks, adhere to data privacy requests, and maintain high efficacy throughout their operational lifecycle can be created by directly incorporating the concepts and mechanisms of machine unlearning into the architecture of malicious document detectors.

III.SURVEY

Sr. no.	Name of paper	Author Name(s)	Advantage or Outcome	Limitation
1.	Towards Making Systems Forget with Machine Unlearning	Yinzhi Cao, Junfeng Yang	Fast, general unlearning via **summation form**; asymptotically faster than retraining.	Not applicable to all algorithms (e.g., DNNs); **relies on identifying** polluted data.
2.	Malicious PDF Detection Based on Machine Learning with Enhanced Feature Set	Suleiman Y. Yerima, Abul Bashar, Ghazanfar Latif	Novel **"anomaly-based" features** improved detection accuracy, outperforming prior work.	Static analysis is vulnerable to **advanced obfuscation and** not tested against adversarial attacks.
3.	One-Class SVMs for Document Classification	Larry M. Manevitz, Malik Yousef	**One-Class SVM** is effective for classification using only positive data.	Highly **sensitive to data representation** and kernel choice, lacking robustness.
4.	Towards Adversarial Malware Detection: Lessons Learned from PDF-based Attacks	Davide Maiorca, Battista Biggio, Giorgio Giacinto	Comprehensive **taxonomy of attacks** on PDF detectors; identified parser vulnerabilities.	Attackers face **inverse feature-mapping problems**; no single detector is universally robust.
5.	Static Detection of Malicious JavaScript-Bearing PDF Documents	Pavel Laskov, Nedim Šrndić	**PJScan** is a highly efficient (<50 ms) static detector suitable for large-scale use.	Vulnerable to **code obfuscation** and fetching from other locations; high lab false positive rate.
6.	The Security of Machine Learning	Marco Barreno, Blaine Nelson, Anthony D. Joseph, J.D. Tygar	Foundational **attack taxonomy** (Influence, Violation, Specificity) provides a common language.	The proposed **RONI defence** is less effective against targeted attacks and computationally costly.
7.	A Survey of Machine Unlearning	Thanh Tam Nguyen et al.	The proposed **RONI defence** is less effective against targeted attacks and computationally costly.	No single algorithm meets all design requirements; it **lacks unified benchmarking** standards.
8.	Poisoning Attacks against Support Vector Machines	Battista Biggio, Blaine Nelson, Pavel Laskov	Practical poisoning attack against SVMs using **kernelised gradient ascent**.	Assumes **significant attacker knowledge** (algorithm, validation set); the attacker controls labels.
9.	A Survey of Security and Privacy Issues of Machine Unlearning Aobo Chen et	Aobo Chen et al.	First survey on **security/privacy vulnerabilities** of the unlearning process itself.	First survey on **security/privacy vulnerabilities** of the unlearning process itself.
10.	DYNFRS: An Efficient Framework for Machine Unlearning in Random Forests	Shurong Wang et al.	**DYNFRS** enables efficient, exact unlearning for random forests via subsampling and lazy tagging.	Performance relies on ERT robustness; **optimal parameters are dataset-specific**.

IV. Flow Diagram



Descriptions for Data Flow Diagram Blocks

1. PDF Dataset

This is the foundational data source for the entire system. It consists of a large, labeled collection of PDF files, partitioned into two distinct classes: a set of known malicious documents and a set of known benign documents. This corpus is essential for training, testing, and validating the machine learning model's ability to accurately distinguish between harmful and safe files.

2. Extract Features from PDF

In this stage, a static analysis process is performed on each PDF file to convert its structural and content-based properties into a numerical format. A specialized parser (e.g., PyMuPDF, PDFiD) inspects the raw file to extract a predefined set of features, such as the count of specific keywords (/JS, /OpenAction), file size, number of pages, presence of embedded files, and characteristics of embedded scripts. The output is a feature vector for each PDF, which serves as the input for the subsequent machine learning stages.

3. Data Preprocessing

This block involves cleaning and preparing the extracted feature vectors for model training. The tasks performed here include handling missing values (e.g., through imputation), normalizing numerical features to a common scale (e.g., min-max scaling), and encoding any categorical features. The goal is to ensure the data is in a consistent and optimal format, which improves the convergence and performance of the machine learning algorithm.

4. EDA (Exploratory Data Analysis)

Exploratory Data Analysis is the critical process of investigating the dataset to discover patterns, identify anomalies, and check assumptions with the help of statistical summaries and graphical visualizations. In this context, EDA is used to understand the distribution of different features and analyze the correlations between them. The insights gained from EDA, such as identifying the most discriminative features between malicious and benign files, inform the feature selection process and guide the model building strategy.

5. Training ML Model (One-Class SVM)

This is the core learning phase where the machine learning model is trained. A **One-Class Support Vector Machine (OCSVM)** is used, which is an anomaly detection algorithm particularly suited for this task. It is trained primarily on the feature vectors of the *malicious* PDF samples. The OCSVM learns a decision boundary (a hypersphere in the feature space) that encapsulates the characteristics of "normal" malicious data. The output is a trained model capable of identifying whether a new, unseen PDF fits this profile of maliciousness.

6. Unlearn Selective Samples

This block represents the novel machine unlearning workflow of the project. When specific training samples are identified as "poisonous" or need to be removed (e.g., to simulate recovery from a data poisoning attack), an unlearning request is triggered. An approximate unlearning algorithm is then executed to efficiently update the trained OCSVM model, removing the influence of the selected samples without the need for a full, time-consuming retraining process from scratch. The output is a repaired, updated model.

7. Determine a PDF is malicious/benign.

This is the final inference or prediction stage of the system. An unseen PDF file is processed through the same feature extraction pipeline, and its resulting feature vector is fed into the live (either initially trained or subsequently unlearned) OCSVM model. The model calculates the sample's distance from the learned decision boundary. Based on whether the sample falls inside or outside this boundary, the system classifies the PDF as either malicious (an inlier) or benign (an outlier/anomaly), providing the final security verdict.

V. CONCLUSION

This project successfully introduced and demonstrated a novel method for improving the security of machine learning systems specifically designed for malicious PDF detection. We confirmed that One-Class SVM classifiers, despite their effectiveness in identifying malicious documents, are susceptible to data poisoning attacks, which can severely compromise their performance. The traditional approach of complete model retraining is not a viable solution for maintaining system integrity due to its prohibitive computational cost.

This project's primary achievement was the successful implementation and validation of an **approximate machine unlearning** mechanism. This mechanism serves as a direct and efficient defense against data poisoning attacks. We demonstrated that in a simulated data poisoning scenario, our unlearning workflow effectively eliminated the influence of malicious training samples from the compromised model. Our evaluation confirmed that this approach restores the detector's accuracy to nearly that of a perfectly retrained model, but in significantly less time. This validates machine unlearning as a viable, efficient, and essential tool for developing resilient and trustworthy security systems in adversarial environments, moving it beyond a theoretical concept.

VI. Future Scope

While this project establishes a strong proof-of-concept, there are several promising directions for future research and development:

Expanding Unlearning Techniques:

Implement and compare the efficiency of other unlearning techniques, including exact methods like the SISA (Sharded, Isolated, Sliced, and Aggregated) framework. This is particularly relevant for complex ensemble models such as Random Forests.

Unlearning for Deep Learning Models:

Adapt and test unlearning algorithms specifically designed for Deep Neural Networks (DNNs). This is a critical next step as deep learning becomes more prevalent in malware detection, despite the challenges posed by the non-convex nature of deep learning optimization.

Automating Poisoning Detection:

Integrate an automated detection mechanism into the current workflow, which currently relies on manual identification of the "forget set" by an administrator. This advanced system could monitor model performance for sudden drops in accuracy, flag suspicious data points, and automatically trigger the unlearning process.

Enhancing Robustness Against Unlearning-Based Attacks:

Investigate potential vulnerabilities of the implemented unlearning algorithm and explore defensive strategies. This is crucial given the emerging research on attacks that specifically target the unlearning process, making the model repair process more secure.

Expanding the Feature Set:

Enhance the feature extraction module to combat more sophisticated evasion techniques. This could include features derived from dynamic analysis or more advanced static analysis, such as control-flow graphs of embedded JavaScript.